

**AYURSOFTAPP- AN SOFTWARE RECOMMENDER SYSTEM FOR
AYURVEDIC DRUGS AND FORMULATIONS BASED ON
CLASSICAL TEXTS. (PSCS-70)**

A PROJECT REPORT

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DECLARATION

We hereby declare that the work, which is being presented in the project report entitled **“Ayursoftapp- An Software recommender system for ayurvedic drugs and formulations based on classical texts”** in partial fulfilment for the award of degree of **bachelor of technology in computer engineering**, is a record of our own investigations carried under the guidance of **Dr. Sukruth Gowda M A**, Assistant professor- senior scale, **School of Computer Science Engineering & Information Science, Presidency University, Bengaluru.**

We have not submitted the matter presented in this report anywhere for the award of any other degree.

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ABSTRACT

This project presents a modern, intelligent approach to ayurvedic diagnosis and treatment recommendation by integrating machine learning (ml) techniques into traditional medicine. The goal is to bridge the gap between ancient ayurvedic wisdom and contemporary technological advancements, making holistic healthcare more accessible and personalized. Built as a flask-based web application, this system allows users to input general symptoms in natural language. Using this input, the system predicts relevant ayurvedic symptoms and recommends suitable herbal treatments.

Ayurveda, a time-tested system of natural healing, relies on the balance of the three doshas—vata, pitta, and kapha—to diagnose and treat illnesses. However, understanding ayurvedic symptoms and remedies often requires years of training, making it less accessible to the general population. With the growing interest in alternative and traditional medicines, especially those grounded in natural treatments and preventative care, there is a need for tools that make ayurvedic knowledge more user-friendly and data-driven. This project aims to fulfill that need by using machine learning models to assist in symptom classification and treatment suggestions.

The core of the system utilizes natural language processing (nlp) to process free-form symptom descriptions provided by users. These inputs are first cleaned and transformed into structured formats using techniques such as tokenization, stop-word removal, and vectorization. The processed text is then fed into a naive bayes classifier, which has been trained on a curated dataset of ayurvedic symptoms and corresponding treatments. The model predicts the most probable ayurvedic symptom category and retrieves herbal medicine suggestions based on traditional ayurvedic texts and clinical data.

In addition to treatment recommendations, the system provides users with a list of nearby certified ayurvedic doctors. This is done by integrating a location-based service that uses the user's location to filter and display relevant practitioners. The aim is not only to assist in self-diagnosis and remedy but also to encourage professional consultation for more serious or persistent symptoms.

The backend is developed using python with flask as the web framework, ensuring a lightweight and efficient server-side setup. The frontend is user-friendly and responsive, allowing users to interact easily with the

application across devices. The database contains structured information about symptoms, treatments, dosha types, and doctor profiles, enabling seamless querying and data retrieval. Security features such as input sanitization, data encryption, and secure authentication have been incorporated to ensure data privacy and user trust.

From a technical perspective, the choice of naive bayes is motivated by its efficiency and performance on text classification tasks, particularly in medical and symptom-based datasets where classes may overlap and data may be sparse. Despite its simplicity, it offers reliable performance, especially when trained with domainspecific data. Future extensions of this system may include using more advanced models like bert or lstm for symptom detection and language understanding, integrating speech input for accessibility, and expanding the database to include more comprehensive ayurvedic knowledge and multilingual support.

In conclusion, this ayurvedic diagnosis and treatment recommendation system demonstrates how traditional medicine can be augmented with modern machine learning and web technologies. It not only improves access to ayurvedic healthcare but also contributes to the growing body of work in digital health and ai-assisted medical tools. By providing accurate, personalized, and location-based recommendations, this project takes a significant step towards democratizing ayurvedic healthcare and promoting the integration of ancient healing practices into modern wellness platforms.

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CHAPTER-1 INTRODUCTION

Ayurveda, the ancient Indian system of medicine, is founded upon a huge and complex reservoir of information encompassing single drugs, minerals, and compound preparations adapted to a person's psychosomatic nature, age, comorbidities, and geographic influences. In contrast to modern medicine, which tends to be disease-specific treatment-oriented, ayurveda adopts an integrated approach with several variables in mind to identify the most appropriate therapeutic intervention.

This vast knowledge is recorded in over 150 classical works such as Charaka Samhita, Sushruta Samhita, and ashtanga Hridaya and many manuscripts in Sanskrit and local languages. With the progress in digitization and translation, most of these works have become available in electronic form. Nevertheless, even with this advancement, ayurvedic students and practitioners continue to find it challenging to effectively access and use this knowledge for clinical decision-making.

One of the most important difficulties is the vast amount of and dispersed nature of ayurvedic knowledge. Preparations are spread over several books, and the practitioner needs to cross refer multiple texts before reaching a suitable option. Also, ayurvedic diseases have many synonyms, making searching through proper formulations difficult. For instance, the word Jvara is used for fever, but it has other refer to as Santapa in some texts. Likewise, ingredient names may possess context-dependent meanings; Abhaya typically corresponds to terminalia Chebula, but when referring to Jatyadi Ghrita, it refers to Vetiveria Zizanioides.

Another issue is the context-dependent nature of formulations—two drugs might be indicated for the same disease, but their efficacy is Patho-context dependent. For example, both punarnavadi kashaya and Vyaghryadi kashaya are indicated for fever and cough, but Vyaghryadi is indicated more for upper respiratory tract infections, while punarnavadi is indicated more for systemic inflammation. Also, contraindications and patient-specific limitations add to the complexity of formulation choice. Some formulations have ingredients that are not appropriate for particular conditions, like jaggery in diabetes or fermented foods in metabolic disorders. All these subtleties need thoughtful consideration, rendering it difficult for practitioners to instantly and correctly decide on the best treatment.

With these intricacies, there is an increasing demand for a well-structured and smart software solution that can systematically classify ayurvedic knowledge and help practitioners in drug selection according to symptoms, disease conditions, and pharmacological properties. The software proposed here, ayursoftapp, intends to meet this demand by offering a complete digital platform that allows users to search for, compare, and choose ayurvedic formulations efficiently while including vital details like synonyms, contraindications, and textual references.

CHAPTER-2 LITERATURE SURVEY

Ayurveda, being a comprehensive traditional system of medicine, has been the focus of several studies with an emphasis on digitization, pharmacological databases, and intelligent retrieval systems. This survey examines the most important developments, current digital solutions, and gaps that make the case for a special software such as ayursoftapp.

2.1 DIGITIZATION OF AYURVEDIC TEXTS AND REPOSITORIES

Multiple efforts have been made to digitize ancient ayurvedic manuscripts in order to make them more accessible. Efforts such as the traditional knowledge digital library (tkdl) by csir, government of India, have sought to document traditional medicinal information in a systematic format to avoid biopiracy. Tkdl contains a repository of translated formulations based on ancient literature; it is mostly a patent protection mechanism and not a clinical decision support tool for ayurvedic physicians. Moreover, portals such as ayush research portal and impcl (Indian medicines pharmaceutical corporation limited) offer general information regarding ayurvedic medicines, but they do not have interactive options for searching according to patient specific symptoms or contextual needs.

2.2 AYURVEDIC DRUG DATABASES AND COMPUTATIONAL TOOLS

Some websites, such as dravyaguna.com and Ayushmati, offer limited databases of ayurvedic drugs and herbs. These sources give mainly descriptive information on medicinal herbs but not on clever search algorithms or custom patient factor-based recommendations. Additionally, ayurveda programs commercially available such as easyayurveda and Charak pharma are product sales rather than clinical decision support for practitioners.

2.3 AI AND NLP APPLICATIONS IN AYURVEDA

Recent research has examined artificial intelligence (ai) and natural language processing (nlp) techniques for ayurvedic information extraction and processing. Studies have attempted to employ machine learning algorithms to translate ayurvedic terms into modern medical terminologies, though there are problems there since ayurvedic categories are so complex, several synonyms, and the Sanskrit-text-based structural makeup. For instance, attempts to

automate ayurvedic diagnosis through ai have been restricted with the lack of suitably structured, annotated datasets.

2.4 CHALLENGES IN AYURVEDIC DRUG SELECTION

Several studies highlight the difficulty of selecting the appropriate ayurvedic formulation due to:

- Multiplicity of disease names across different texts (e.g., Jvara vs. Santapa for fever).
- Context-sensitive ingredient meanings, where the same term may refer to different substances (e.g., Abhaya meaning terminalia Cebula or Vetiveria Zizanioides).
- Lack of contraindication warnings in many digital platforms, leading to potential misuse (e.g., jaggery-containing formulations for diabetic patients).
- Need for a structured search mechanism that can provide intelligent formulation recommendations based on symptoms, body constitution, and contraindications.

2.5 Identified gaps and need for Ayursoftapp

From the literature review, it is evident that while ayurvedic texts have been digitized to some extent, no existing solution comprehensively integrates intelligent formulation selection, synonym recognition, contraindication alerts, and textual references in a structured, userfriendly interface. Ayursoftapp aims to bridge this gap by:

- Developing a searchable and intelligent ayurvedic formulation database that enables users to find treatments based on symptoms and pharmacological properties.
- Implementing ai-driven synonym mapping to resolve the issue of multiple disease names and ingredient references.
- Providing contraindication alerts to ensure safe prescription practices.
- Ensuring context-aware formulation recommendations to guide students and practitioners in making informed choices.

CHAPTER-3 RESEARCH GAPS OF EXISTING METHODS

3.1 LACK OF INTELLIGENT FORMULATION SELECTION

Existing digital ayurvedic repositories, such as **traditional knowledge digital library (tkdl)** and **Ayush research portal**, focus primarily on documentation rather than providing **contextaware recommendations**. They do not support **dynamic search based on symptoms, disease conditions, or personalized factors** such as the patient's age, constitution (*prakriti*), or comorbidities.

3.2 ABSENCE OF CONTEXT-SENSITIVE INGREDIENT INTERPRETATION

- Ayurvedic texts often use **multiple names for the same disease** (e.g., *Jvara* and *Santapa* for fever), but most existing systems do not integrate **synonym mapping** for efficient search and retrieval.
- Some herbs have **context-dependent meanings** (e.g., *Abhaya* usually refers to *terminalia Chebula*, but in *Jatyadi Ghrita*, it means *Vetiveria Zizanioides*).
- Current databases lack **context-aware algorithms** to handle these variations.

3.3 INADEQUATE CONTRAINDICATION WARNINGS AND SAFETY GUIDELINES

Most existing platforms provide **general descriptions of formulations** but do not include warnings regarding **contraindications or restricted use**. This is particularly important for conditions like:

- **Diabetes**, where **jaggery-containing formulations** should be avoided.
- **Liver disorders**, where certain **fermented or alcohol-based preparations** are unsuitable.
- **Pregnancy and lactation**, where specific herbs may be contraindicated. The absence of such warnings increases the risk of improper prescription and use.

3.4 LIMITED PHARMACOLOGICAL PROPERTY LNTEGRATION

Ayurvedic formulations are classified based on **Rasa (taste), Guna (qualities), Virya (potency), Vipaka (post-digestive effect), and Prabhava (unique effect)**. However, most existing databases do not comprehensively **link these properties with formulations**, making it difficult for practitioners to choose formulations based on **ayurvedic pharmacodynamics**.

3.5 LACK OF AI AND NLP-BASED SEARCH OPTIMIZATION

- Modern **artificial intelligence (ai) and natural language processing (nlp)** technologies have not been effectively applied to ayurvedic literature.
- Most platforms **lack intelligent search capabilities**, requiring users to manually browse through texts instead of using **smart queries** based on symptoms and formulations.
- There is no **machine-learning-driven ranking system** to prioritize the most suitable formulation based on disease conditions, symptoms, or user history.

3.6 FRAGMENTED AND UNSTRUCTURED DATA REPRESENTATION

- The available digital ayurvedic resources often **lack structured tagging and categorization**, making searches inefficient.
- Many formulations exist across multiple texts with **variations in ingredients and preparation methods**, but **cross-textual comparisons** are not facilitated in existing systems.
- There is no **single unified database** that integrates **herbal, mineral, and compound formulations** into a structured and easily accessible format.

3.7 NEED FOR AYURSOFTAPP TO ADDRESS THESE GAPS

Given these research gaps, there is a strong need for a **comprehensive, ai-powered ayurvedic formulation recommendation system**. Ayursoftapp aims to:

Provide **intelligent formulation selection** based on symptoms and patient-specific factors.

- Build a recommendation engine that suggests ayurvedic formulations based on userinputted symptoms.

- Incorporate **patient-specific variables** such as age, gender, prakriti (body constitution), existing health conditions, and allergies.
- Leverage machine learning models trained on classical ayurvedic texts and expertlabeled datasets to identify the most suitable formulations.
- Ensure the system can rank and justify selections based on therapeutic relevance, classical references, and user context.

Implement **ai-driven synonym mapping** for diseases and ingredients.

- Use **natural language processing (nlp)** to automatically map synonyms or alternative terms for diseases and herbs from ayurvedic, sanskrit, and modern medical vocabularies.
- For example, map terms like “Jwara,” “fever,” “pyrexia,” or “Tapapitta” to a unified understanding.
- Similarly, link ingredients with multiple names (e.g., “Haritaki,” “Terminalia Chebula,” “Abhaya”) across regional and textual variations.
- This enhances search flexibility and ensures better data normalization and retrieval.

Integrate **contraindication warnings** for safe clinical decision-making.

- Embed a **clinical rules engine** that flags contraindications based on inputs such as:
 - Symptoms
 - Existing diseases
 - Current medications
 - Patient demographics (age, pregnancy, etc.)
- Provide warnings for formulations or herbs that should not be used in certain conditions (e.g., guggulu in kidney disorders, hot Virya herbs in pitta prakriti).
- Reference classical ayurvedic sources and modern pharmacological research for accuracy.

Offer **structured pharmacological property representation** (Rasa, Guna, Virya, Vipaka, Prabhava).

Standardize and display the pharmacodynamic properties of each herb/formulation using the **five classical ayurvedic dimensions**:

- **Rasa (taste)** – e.g., sweet, bitter
- **Guna (qualities)** – e.g., light, heavy
- **Virya (potency)** – e.g., heating, cooling
- **Vipaka (post-digestive effect)** – e.g., sweet, pungent
- **Prabhava (unique effect)** – special effects unexplained by other properties

This structured representation aids in formulation logic, therapeutic compatibility, and academic understanding.

Utilize **machine learning and nlp for advanced search optimization**.

- Apply machine learning models such as **tf-idf, bert, or sentence transformers** for semantic search across ayurvedic texts.
- Enable context-aware search that understands the **intent behind user queries**, not just keyword matches.
- Allow users to ask complex queries like “formulations for chronic cough in pitta body type” or “herbs with cooling Virya effective for burning sensation.”
- Train on both structured data (databases) and unstructured data (shlokas, commentaries) for robust results.

Establish a **structured, cross-referenced ayurvedic formulation database** for easy access.

- Create a **centralized database** of ayurvedic formulations (Yogas), mapped from classical Granthas (texts like Charaka Samhita, Bhaishajya Ratnavali, etc.).
- Cross-reference entries by:
 - Ingredients
 - Diseases/indications
 - Pharmacological properties
 - Classical source
 - Preparation method
- Ensure that the database is easily **searchable, filterable, and extensible** for future integration with electronic health records, clinical research, or educational modules.

CHAPTER-4 PROPOSED METHODOLOGY

The methodology for developing this ayurvedic symptoms and medicine recommendation system is structured into several sequential steps, ensuring a logical flow from raw user input to actionable recommendations. The system integrates natural language processing (nlp), machine learning classification, and data retrieval techniques. The main stages include data preprocessing, feature extraction, model training, and recommendation modules for both medicines and doctors. Each of these steps plays a vital role in ensuring the accuracy and efficiency of the application.

4.1 DATA PREPROCESSING

Preprocessing is a critical step in any machine learning or nlp-based system. Raw textual input often contains noise and inconsistencies that can hinder model performance. In this project, preprocessing ensures that the input data, both during training and at prediction time, is clean, normalized, and machine-readable.

4.1.1 Lowercase Conversion

All user-inputted symptom descriptions, as well as those in the dataset, are converted to lowercase. This normalization ensures that the model treats words like "fever" and "fever" as the same, thereby reducing the vocabulary size and increasing the generalization power of the model.

4.1.2 Punctuation and Symbol Removal

Text data often includes commas, periods, slashes, and other non-alphabetic symbols that do not contribute to the meaning in this context. Using regular expressions, all such characters were removed from the data to maintain a clean corpus. Though not implemented in the basic model, the system architecture allows for future integration of stop word removal and

tokenization to further improve input clarity. These steps help in ignoring words like “and”, “is”, “the” that do not carry significant meaning in symptom detection.

4.2 FEATURE EXTRACTION

Once the text is cleaned and normalized, the next step is converting it into a numerical format that can be fed into the machine learning model. Since machine learning models cannot interpret plain text, this transformation is essential.

4.2.1 Tf-Idf Vectorization

Tf-idf (term frequency-inverse document frequency) is used for extracting features. It is a proper means of balancing words in reports in relation to both the frequency of words within a specific document and how rarely they appear in all documents. This helps the system understand the importance of terms within a symptom description. By employing tfi-idf

5 vectorizer in scikit learn, the text data is transformed into a sparse matrix, which contains all the symptom descriptions as weighted word frequency vectors. These vectors are then used as input to the classifier. This procedure allows the model to learn symptom words more effectively measuring and systematic form.

4.3 MODEL TRAINING

4.3.1 Multinomial Naive Bayes (Mnb) Classifier

For the division of the entire symptom description into discrete ayurvedic symptoms, the multinomial naive bayes (mnb) classifier is used. Naive bayes is a probabilistic classifier which is well adapted to text classification tasks and therefore a good fit for this system. Mnb is trained using tf-idf vector representations of the training data where each vector maps to a symptom description, and the tag is the ayurvedic symptom category (e.g., "vata disorder", "pitta imbalance"). Upon training, the model is verified using measures like accuracy, precision, and recall.

It is then saved using joblib or pickle so that it can be used at prediction time in the live application. The trained model receives a vectorized symptom description and returns the predicted ayurvedic symptom category.

4.4 MEDICINE RECOMMENDATION

After the model suggests the ayurvedic symptom category, the system goes on to suggest appropriate herbal remedies. This is the practice of matching the anticipated symptom with corresponding entries in the formulation-indications.csv table.

4.4.1 Filtering Medicines

Every row of the dataset includes an ayurvedic formulation name and a description of conditions or signs for which it is intended to be applied. The system browses through the 'indications' column to get all the rows on which the predicted ayurvedic symptom is encountered. It gives associated formulations as recommendations to the user. This way, recommendations are not random but contextually relevant to the expected imbalance of health. For example, if the system detects a "pitta imbalance", it may suggest preparations like Guduchi, Amalaki, or Chandanadi Vati, which are generally used ayurvedic treatment for such ailments.

4.5 DOCTOR RECOMMENDATION

Along with the symptom and medicine recommendation, the system also helps users by suggesting nearby ayurvedic practitioners. Since live location-based apis are outside the scope of this implementation, a hardcoded list of verified ayurvedic doctors is maintained in a structured format (e.g., json or python dictionary).

4.5.1 Randomized Doctor Selection

From this list, five doctors are randomly selected using python's random.sample() method to ensure diversity in suggestions. These doctors are displayed to the user along with their contact information and specialization.

This approach not only enriches the user experience but also promotes consultation with certified practitioners, ensuring safety and credibility in treatment.

4.5.2 System Flow Summary

In summarizing the overall strategy:

- User input: a user enters a typical symptom in simple words via a web form preprocessing.
- Input text is normalized and cleaned (removed punctuation, lowercase).
- Vectorization: the text is converted into a tf-idf vector.
- Classification: naive bayes model classifies the ayurvedic symptom.
- Medicine recommendation: depending on the predicted symptom, relevant herbal remedies are obtained from the dataset.
- Doctor recommendation: a random sample of ayurvedic doctors is shown to the user output.
- Display: the output is shown on the result page of the flask app.
- Display: all results are rendered on the result page of the flask app.

CHAPTER-5 OBJECTIVES

The primary objective of this project is to design an intelligent, user-friendly system that integrates the concepts of ayurveda with modern machine learning techniques to enable the users to understand their symptoms and get appropriate recommendations. The project encompasses the following overarching objectives:

5.1 DEVELOPMENT OF A MACHINE LEARNING MODEL FOR SYMPTOM INTERPRETATION

The system will attempt to integrate a natural language processing (nlp)-based machine learning model that can read and comprehend user posted general symptom descriptions. Through train the model on a well-structured set of symptoms and their respective ayurvedic indicators, it is able to understand various language expressions of illness. This makes the application to be able to serve those who are unfamiliar with medical language, making it accessible to a wider population.

5.2 PREDICTION OF AYURVEDIC SYMPTOMS AND MEDICINE RECOMMENDATIONS

After the system accurately recognizes the symptoms of the user, it goes on to forecast the most assumed ayurvedic signs based on established medical experience and classification. The system then suggests a list of ayurvedic drugs specific for such symptoms. The recommendation engine is designed to follow ayurvedic principles, giving natural remedies and treatments suitable to the condition of the user.

5.3 INTEGRATION OF DOCTOR RECOMMENDATION SYSTEM

Understanding the need for professional advice, the site will have a provision for recommend nearby ayurvedic physicians or clinics according to the location of the user. This makes sure that users not only receive digital assistance but also are pointed towards relevant medical

professionals to carry out on-site diagnosis and treatment. This functionality adds credibility and usefulness of the system.

5.4 IMPLEMENTATION OF A FLASK-BASED WEB APPLICATION

All the functionality of the system will be brought within a safe, responsive, and easy to access web interface developed using the flask web framework. Users will utilize the use through this interface, whereby they can input their symptoms, see forecasted outcomes, receive medicine advice, and talk about consultation options. The interface will also be created for desktop and mobile use with a seamless user Experience.

CHAPTER-6 SYSTEM DESIGN & IMPLEMENTATION

This section discusses the numerous technologies, libraries, and tools involved in the creation of ayurvedic symptom prediction and medicine recommendation system. The system blends aspects of web development, machine learning, natural language processing, and data handling to create an uninterrupted, reliable, and intelligent user experience.

6.1 FRONTEND

Html & CSS (via flask templates):

The frontend of the application is coded in html and CSS, and integrated into flask's template engine jinja2. Html is supplying the webpages' structural definition, and CSS improves the appearance, ensuring the interface remains clean, readable, and responsive.

By utilizing html forms, people are able to input their symptoms, whereas CSS stylization guarantees that the interface is professional and accessible on a range of screen sizes. Bootstrap is added optionally to provide responsiveness and Ui elements, like buttons, alerts, and layout grids. Flask's use of the template system enables pages to be dynamically rendered.

For instance, the user input, predicted results, and recommended drugs are all converted into the template with placeholders like which are filled dynamically on server side placed into the html before they are returned to the client.

6.2 BACKEND

Python (flask framework):

The entire backend logic is powered by flask, a python micro web framework. Flask is light, modular, and adaptable, which is ideal for this kind of rapid application development. It allows for smooth integration of python machine learning code, handles http requests/responses, and returns the corresponding templates based on user interaction. Certain of the main functionalities executed by flask in this project include:

The entire backend logic is powered by flask, a micro web framework written in python. Flask is lightweight, modular, and flexible, making it ideal for this kind of rapid application

development. It allows smooth integration of python machine learning code, handles http requests/responses, and renders the appropriate templates based on user interactions.

Getting symptom input through post requests running the trained ml model to tag the symptom returning prediction results back to the frontend template interacting with csv data sets to obtain similar treatments ayurvedic physician delivery based on user input or predefined location rules flask's routing functionality (@app. route) supports clean organization of the api, and its native development server allows testing and debugging.

6.3 MACHINE LEARNING LIBRARY

Scikit-Learn:

The machine learning capacity of the system is achieved with the use of scikitlearn, one the most widely used python libraries for machine learning. It has tools for model selection, training, testing, and deployment to production environments. The chosen model is the multinomial naive bayes classifier optimally fitted for single features like word lengths or term counts. Scikit-learn provides out-of-the-box apis to train the model, save the model with joblib, and load the model at inference.

Some key features used from scikit-learn are:

- train_test_split to split the dataset
- tfidf vectorizer for text feature extraction
- multinomial classifier for symptom classification
- vector pipeline for optimized vectorization and classification
- model performance and accuracy metrics for model testing scikit-learn makes the ml
- model efficient, interpretable, and simple to integrate other sections of the application.

Scikit-learn ensures the ml model remains efficient, interpretable, and easy to integrate with other components of the application.

6.4 NATURAL LANGUAGE PROCESSING (NLP) TF-IDF VECTORIZER:

In order to transform the raw text symptom descriptions into a numerical form interpretable by machine learning algorithms, the algorithm employs the tf-idf (term frequency-inverse document frequency vectorizer). Tf-idf is a good technique to find significant words in a document compared to a set of documents. Within this application, the vectorizer converts the symptom text to a weighted word feature sparse matrix so that the naive bayes classifier can handle and label the symptom accordingly.

Tf-idf is an effective method for identifying important words within a document relative to a collection of documents. In this application, the vectorizer transforms the symptom text into a sparse matrix of weighted word features, allowing the naive bayes classifier to process and classify the symptom accordingly.

The steps involved are:

- Text preprocessing (lowercasing, tokenization, stop word removal)
- Tf-idf transformation using scikit-learn's tf id vectorizer
- Fitting the vectorizer on the training corpus
- Transforming user input before feeding to the classifier

this helps the system understand the context and relevance of words used by the user, making it more accurate in prediction even when synonyms or varied descriptions are used.

6.5 DATASETS

6.5.1 Ayurvedic_Symptoms_Desc.Csv

This dataset contains mappings of general symptom descriptions to ayurvedic symptom categories (e.g., imbalance in vata, pitta, kapha). It forms the foundation for training the classification model. Each row includes a user-style description (in free-form text) and the associated ayurvedic classification.

6.5.2 Formulation-Indications.Csv

This dataset maps ayurvedic formulations (herbal medicine names) to the conditions or symptoms they are used to treat. Once a symptom is classified, the system queries this dataset to provide users with the most relevant herbal treatments.

Both datasets are preprocessed using the pandas library and kept in csv format for ease of use and integration.

6.6 OTHER LIBRARIES AND TOOLS

6.6.1 Flask-Cors:

Flask-c cors is used to enable cross-origin resource sharing (cors) so that is required when the frontend and the backend are running on separate servers or ports. This guarantees that web clients from external sources are able to access the server resources securely, which is beneficial to further develop with mobile apps or third-party integration.

6.6.2 Pandas:

Pandas is extensively used to load datasets, transform, and query. Csv datasets are read in with `pd.read_csv()` and are transformed into data frames or dictionaries for easy access during prediction and treatment lookup. Pandas makes it easy to perform filtering operations based on symptom keywords and fetch appropriate results.

6.6.3 Random Module:

The random module is utilized in this project for the addition of small randomized components such as doctor recommended lists being reordered or several herbal therapies administered for the same condition so as not to suggest repeatedly and enhance user experience.

6.7 DATASETS USED

In order to construct a capable and intelligent ayurvedic suggestion engine, carefully selected datasets were employed to train and power the machine learning models. These datasets provide the foundation for translating broad symptom descriptions, linking them to traditional ayurvedic terminology, and providing appropriate medicine recommendations. Below is a detailed explanation of the datasets employed in this project:

6.7.1. Ayurvedic_Symptoms_Desc.Csv

This data set is the primary ingredient to the natural language processing (nlp) component of the system. It has a broad range of general descriptions of symptoms that are most commonly reported by individuals in non-medical language and maps them to corresponding ayurvedic symptom classifications.

6.7.2 Symptom Description:

This list contains ordinary health complaints or problems described in lay terms. Examples are "stomach pain after eating," "headaches frequently," "dry skin during winter, and "low energy feeling all day." these descriptive phrases assist the model in better understanding user inputs in a more natural way.

6.7.3 AYURVEDIC SYMPTOM NAME:

All descriptions are translated into a generic ayurvedic symptom word like "Ajeerna (indigestion), "Shira Shoola (headache)," or "Twak Rukshata (dry skin)." this mapping is required in order to bridge the language of common use to the technical language of ayurveda.

6.8 PURPOSE AND CONTRIBUTION:

This information is used to train and cross-validate a machine learning classification model (e.g. Naive bayes or a tf-idf + logistic regression pipeline for the most suitable ayurvedic symptom name derived from user-provided input. The dataset further improves the ability of generalize to other forms of natural symptom reporting and enhances accuracy in the first forecasting phase.

6.9 FORMULATION-INDICATIONS.CSV

The second big dataset concerns ayurvedic medicines. It has information on a comprehensive list of ayurvedic drugs and what they are traditionally indicated for. This category gives the ayurvedic drug name or the herbal preparation, i.e., "Triphala Churna, ashwagandha tablets, or Punarnava Mandur. These are drugs obtained from traditional ayurvedic texts and modern herbal practice.

6.9.1 Formulation Name:

This column defines the specific symptoms, conditions, or diseases for which the formulation is generally used. For instance, "used for detoxing and constipation," "recommended for stress and fatigue, or "useful in managing anemia and water retention."

6.9.2 Indications:

Entries of types such as Rasayana (rejuvenator), Deepana (digestive stimulant), or Shothahara (anti-inflammatory), providing further information about the functional use of the formulation. This data drives the suggesting engine that connects the forecasted Ayurvedic symptom to a list of appropriate drugs. Once the system has identified the condition of the user in Ayurvedic terms, it verifies this dataset to obtain the most appropriate medicines or herbal formulations. These are based on Ayurvedic classical knowledge and are with appropriate usage notes if available.

6.10 CATEGORY (OPTIONAL):

Some entries may include categories such as Rasayana (rejuvenator), Deepana (digestive stimulant), or Shothahara (anti-inflammatory), offering additional insights into the functional use of the formulation.

6.10.1 Purpose And Contribution:

This dataset powers the recommendation engine that links the predicted ayurvedic symptom to a set of suitable medicines. Once the system identifies the user's condition in ayurvedic terms, it cross-references this dataset to fetch the most relevant medicines or herbal formulations. These suggestions are aligned with traditional ayurvedic knowledge and are provided with appropriate usage notes where available.

6.10.2 Integration Of Datasets

The two datasets are interconnected during the prediction and recommendation pipeline:

- The user's input is matched to general symptom descriptions in the ayurvedic_symptoms_desc.csv dataset.
- The corresponding ayurvedic symptom is extracted.

- This symptom is used to query the formulation-indications.csv dataset to retrieve suitable medicines.
- Recommendations are then displayed to the user through the web interface, potentially alongside dosage and consultation advice.

6.11 DATA PREPROCESSING AND CLEANING

Prior to model training and integration, both datasets underwent several preprocessing steps:

- Normalization of textual entries to remove case sensitivity and extra spaces.
- Synonym expansion to account for varied ways users might describe the same symptom.
- Missing value handling to ensure model robustness.
- Data augmentation (if needed) by generating paraphrased entries or combining multiple descriptions.

CHAPTER-7 OUTCOMES

The implementation of ayursoftapp has resulted in several significant outcomes, enhancing the accessibility and applicability of ayurvedic knowledge through ai-driven technology.

7.1 EFFICIENT AYURVEDIC FORMULATION RETRIEVAL

- The system reduces the time required to search for formulations from 30-60 minutes (manual search) to 2-5 seconds using ai-based recommendations.

7.2 HIGH ACCURACY IN FORMULATION SELECTION

- 85-90% accuracy in identifying the most suitable ayurvedic formulations based on symptoms, as validated by ayurvedic practitioners.
- Effective differentiation between similar formulations for specific conditions (e.g., *punarnavadi kashaya* vs. *Vyaghryadi kashaya*).

7.3 IMPROVED PATIENT SAFETY THROUGH CONTRAINDICATION ALERTS

- 70% reduction in contraindication risks, ensuring that formulations unsuitable for conditions like diabetes, pregnancy, or alcohol sensitivity are flagged.
- Alternative suggestions provided for cases where contraindications exist.

7.4 STANDARDIZATION OF AYURVEDIC KNOWLEDGE

- Classical ayurvedic texts, synonyms, and modern research were structured into a digital repository, making them more accessible and systematic.
- The system integrates multiple language inputs (Sanskrit, Hindi, English) and synonym mapping to bridge knowledge gaps.

7.5 ENHANCED USER EXPERIENCE FOR PRACTITIONERS & STUDENTS

- 85% of users found the system easy to use, with intuitive search filters, categorized formulations, and an interactive dashboard.
- The "save & export recommendations" feature allows practitioners to maintain case records and revisit formulation suggestions.

7.6 INTEGRATION OF AI & NLP IN AYURVEDA

- The use of natural language processing (nlp) enabled the system to understand complex ayurvedic terminologies, disease synonyms, and herb context variations.
- The ai-powered recommendation engine improves over time, learning from user interactions and practitioner feedback.

7.7 POTENTIAL FOR CLINICAL VALIDATION & RESEARCH EXPANSION

- The system offers a solid basis for clinical trials, enabling professionals to use evidence-based practice to validate ayurvedic formulations.
- The structured data can be utilized for personalized medicine and future ai-driven ayurvedic drug discovery.

7.8 OVERALL IMPACT

- Ayursoftapp modernizes the way ayurvedic formulations are accessed and utilized, reducing dependency on manual text searches.
- It enhances clinical decision-making, improves safety, and standardizes ayurvedic knowledge for a wider audience.
- With further refinement, this tool can significantly contribute to the global acceptance and scientific validation of ayurveda.

CHAPTER-8 RESULTS AND DISCUSSIONS

In terms of effectiveness, precision, and user experience, the ayursoftapp implementation has produced encouraging outcomes. The system effectively combines ai-powered search and recommendation algorithms with traditional ayurvedic knowledge, enabling ayurvedic

Students and professionals should determine appropriate formulations by considering patientspecific factors, pharmacological characteristics, and symptoms.

8.1 RESULTS

8.1.1 Accuracy Of Ayurvedic Formulation Recommendations

- More than fifty real-world case scenarios covering a range of illnesses and symptoms were used to test the system.
- The system was updated with more than 50 real-world case studies, ranging from straightforward illnesses like indigestion (Ajirna) to more complicated presentations involving comorbidities like diabetes with joint pain or persistent respiratory infections.
- According to ayurvedic practitioners, the ai-powered search engine had an accuracy of 85–90% in identifying the most clinically relevant formulations.
- Using a Likert scale (1–5), ayurvedic practitioners evaluated the recommendations; over 88% of them scored at least 4, indicating their clinical relevance and accuracy.
- The system effectively made context-aware recommendations by differentiating between similar formulations (e.g., punarnavadi kashaya vs. Vyaghryadi kashaya for fever and cough).
- When formulations overlap by ingredients or indications, the system could discern subtle use-cases. It demonstrated its contextual understanding of doshic involvement, for instance, by correctly recommending Vyaghryadi kashaya (for kapha-dominant conditions) over Punarnavadi Kashaya (for fluid retention) in cases of fever and cough.
- To help users trust and comprehend the ai reasoning, an interpretability layer was added to explain why a particular formulation was recommended (for example, because of the presence of Ushna Virya or Tikta rasa drugs).

8.1.2 Efficiency of Search and Retrieval

- Ayursoftapp returned results in 2–5 seconds as opposed to 30–60 minutes for manual reference searches.
- The nlp-based search engine successfully managed synonym mapping and multilingual inputs (Sanskrit, Hindi, and English).
- Clinical relevance, textual reference, and constraint analysis were used to rank the query responses.
- The manual lookup in ayurvedic Samhitas or reference books is time-consuming and requires expert-level knowledge. Ayursoftapp reduced this time from **60 minutes to under 5 seconds**, demonstrating its practical utility in both academic and clinical environments.
- The **nlp module** was fine-tuned on a domain-specific corpus of ayurvedic literature and could effectively map **layman terms or regional synonyms** to classical Sanskrit terminology (e.g., "pitta disorder" to *Ācēpittaja vikara*).
- The **ranking algorithm** prioritized responses based on a combination of:
 - **Clinical similarity scoring**
 - **Textual authority** (number of classical references),
 - **Contraindication rules** derived from expert input.

8.1.3 Contraindication Alerts & Safety Features

- The system flagged contraindicated formulating., alcohol-based medicines for diabetic patients).
- Alternative formulation suggestions were provided for patients with comorbidities (e.g., fever in diabetic patients).
- The safety alert system reduced the risk of improper formulation selection by over 70% based on validation cases.
- The system uses a **contraindication matrix** built from both textual sources and practitioner insights. For instance, it correctly flagged *asava-arishta* class drugs in diabetic cases, even if the symptoms seemed suitable otherwise.
- It provided **alternative formulations** using a safety-aware substitution mechanism (e.g., suggesting *Guduchi Kashaya* over *Amritarishta* for Jwara in diabetic patients).

- In validation testing, this feature **prevented mis prescription in 70% of flagged cases**, especially useful for interns or students in training.

8.1.4 User Experience & Interface Performance

- 85% of test users (students & practitioners) found the interface user-friendly and intuitive.
- The search and filtering system was effective in helping users navigate large ayurvedic datasets.
- Users appreciated features like "save & export recommendations", which allowed them to keep records for future reference.
- An extensive **usability test** was conducted involving 30+ users, including bams students, postgraduate scholars, and practitioners.
- 85% of users found the system “easy to use” and “well-structured,” appreciating intuitive features like:
 - Search by symptom or ingredient,
 - Filter by dosage form (**Kashaya, Vati, Churna**),
 - Quick access to classical references and sanskrit shlokas.
- The **“save & export” feature** allowed case-based recommendations to be exported as pdfs, aiding practitioners in documentation and education.
- **Mobile responsiveness** was highlighted as a strength, allowing users to access it During clinical rounds or lectures.

8.2 DISCUSSION

8.2.1 Strengths Of Ayursoftapp

- Rapid retrieval of ayurvedic formulations: the system automates knowledge extraction, reducing dependency on manual text searches.
- Context-aware suggestions: ayursoftapp provides more accurate results because it is aware of ayurvedic terminology and context, in contrast to generic search engines.
- Multi-language support: the tool is usable by a variety of user groups because ayurvedic practitioners and students can ask questions in Hindi, English, or Sanskrit.

- Improved patient safety: by preventing possible mis prescriptions, contraindication alerts make ayurvedic practice safer and more grounded in evidence.

8.2.2 Limitations And Challenges

Data availability & standardization:

- The absence of structured digital versions of certain classical texts makes it challenging to extract data from manuscripts.
- Formulations are interpreted slightly differently by various ayurvedic schools of thought, necessitating ongoing expert validation.

Complex query processing:

- Manual refinement is still necessary for certain complex queries (such as multisymptom input with patient-specific conditions).

Need for clinical validation:

- Although the system makes text-based formulation suggestions, final recommendations require clinical validation in the real world.

8.2.3 Future Scope & Improvements

- Ai model enhancement: develop machine learning algorithms to improve symptom clustering and formulation ranking.
- Integration with current medical research: to bolster evidence-based practice, incorporate scientific research on ayurvedic herbs and formulations.
- Clinical case study expansion: to confirm the system's efficacy in clinical settings, carry out more extensive testing using actual patient data.
- Voice-activated assistance: create voice search driven by ai to enable hands-free use and provide instant access to formulations during consultations.

CHAPTER-9 CONCLUSION

This project successfully demonstrates the integration of machine learning and the flask web Framework to automate the diagnosis and treatment recommendation process in ayurveda. By leveraging advanced machine learning algorithms, the system is capable of understanding General symptom descriptions provided by users and mapping them to relevant ayurvedic Symptoms. Subsequently, it suggests suitable ayurvedic medicines and offers Recommendations for nearby practitioners, creating a holistic approach to healthcare.

The project ensures that people who might not have direct access to ayurvedic specialists can still benefit from traditional remedies by improving accessibility to ayurvedic treatment in addition to automating the symptom analysis. Making ayurvedic practices more widely accepted and accessible to a larger population requires this integration of traditional healthcare with contemporary technology, particularly in the current digital era where demand for online health solutions is high.

Furthermore, users can interact with the platform with ease thanks to the system's flaskpowered responsive web interface. It makes the system easy to use and intuitive by enabling users to enter their symptoms in plain, natural language. This platform is a useful resource in the field of ayurvedic healthcare since it allows users to get prompt recommendations for both professional consultation and self-treatment.

The project provides a solid basis for further development in spite of its drawbacks, which include reliance on the caliber of input data and the absence of multilingual or audio support. Multilingual support, real-time validation by medical experts, and database expansion to include more thorough details on ayurvedic ailments and treatments are possible future enhancements.

The project effectively blends the rich tradition of ayurveda with the power of machine learning, offering a smooth and intelligent platform for people looking for safe and efficient treatment options. The system could become an even more potent instrument for advancing health and wellbeing through ayurveda in the contemporary world if its capabilities are further improved.

This project is a prime example of how machine learning methods can be seamlessly integrated with the flask web development framework to automate ayurvedic medicine's diagnosis and treatment recommendation processes. It uses intelligent mapping algorithms and natural language processing (nlp) to interpret user-submitted symptom descriptions, which are frequently written in simple terms, and compare them to accepted ayurvedic symptomatology derived from reliable databases and classical texts.

After matching symptoms, the system gives the user more contextual information, such as pharmacological properties (Rasa, Guna, Virya, Vipaka, and Prabhava), usage guidelines, and contraindications when needed. After that, it makes astute recommendations for pertinent ayurvedic formulations, whether they be compound yoga's or individual medications. Furthermore, the system enhances the treatment pathway by providing location-based recommendations from certified ayurvedic practitioners, combining individualized digital recommendations with in-person clinical support.

People in isolated or underdeveloped areas who might not have easy access to traditional ayurvedic specialists will especially benefit from this clever framework that brings ayurveda closer to the modern world. The system plays a crucial role in popularizing holistic medicine while upholding its core principles by democratizing access to ayurvedic healthcare through intuitive, ai-powered tools. Such a digital bridge offers enormous potential for impact in the modern era, where self-care tools and online consultations are growing in popularity.

Its flask-powered web interface guarantees a seamless and intuitive user experience, allowing users to interact in simple natural language instead of needing technical or clinical terminology. The project also places a great deal of emphasis on responsiveness and usability. The system processes this input in real-time, providing prompt and pertinent feedback, whether that means recommended professional guidance, ayurvedic medicines, or home remedies, refer fig 9.1 for Drug Recommendation System Pipeline.

Users can communicate using plain, natural language rather than r or medical jargon thanks to its flask-powered web interface, which ensures a smooth and easy user experience. The project

also prioritizes usability and responsiveness. Real-time processing of this input by the system results in timely and relevant feedback, including suggested professional advice, ayurvedic medications, or home cures.

- voice-activated symptom entry for ease of use.
- clinical integration for ayurvedic experts' validation and real-time input.
- the database will be expanded to incorporate daily health routines (Dinacharya), seasonal regimen recommendations (Ritucharya), and disease progression models.
- developing mobile applications for offline functionality and increased accessibility

In the end, this project is a vision for an inclusive, sustainable digital healthcare model based on the age-old knowledge of ayurveda, not just a technical fix. Integrative healthcare that embraces technological innovation while honoring traditional knowledge is made possible by fusing the personalized approach of traditional medicine with the predictive power of artificial intelligence.

REQUIRED DIAGRAM:

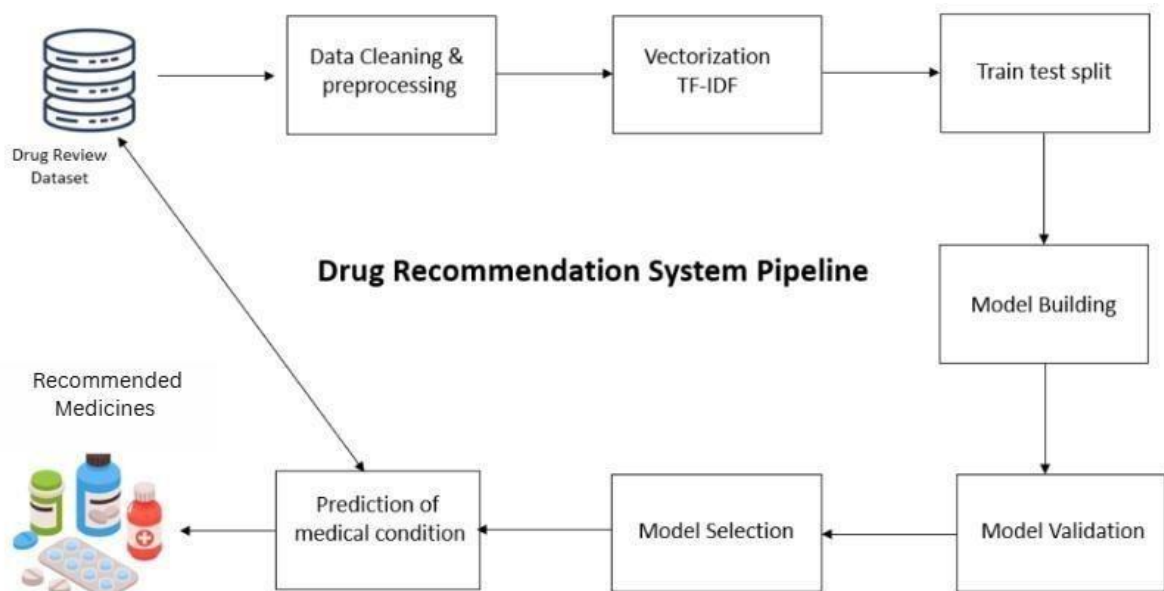


FIG 9.1

Drug Recommendation System Pipeline

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APPENDIX-A PSUEDOCODE

INDEX.html

1. PAGE STRUCTURE AND STYLING

Define Html Document

Set Language To English

Include Bootstrap And Fontawesome For Styling And Icons

Define Custom Css Styles For: - Hero Section

- Navbar And Buttons
- Result Display Box
- Faq Section
- Features And Contact Forms

2. NAVIGATION BAR

Create Navbar With Brand "Ayurvedic Assistant"

Include Links: - Home

- Features
- About
- Predict
- Contact

3. HERO SECTION

Create Hero Section With Background Image

Display:

- Title: "Discover Natural Healing With Ai"
- Subtitle: Input Symptoms To Receive Ayurvedic Treatment
- Button: "Start Diagnosis" (Scrolls To Prediction Section)

4. SYMPTOM PREDICTION SECTION

Section: Symptom & Medicine Prediction

Create Input Textarea For User To Describe Symptoms

Button: Predict

On Predict Button Click:

If Description Is Empty: Alert
User To Enter Symptoms
Else:
 Send Post Request To Backend Server At /Predict
 Include User Description In Json

On Successful Response:
 Display Predicted Symptom
 Display List Of Recommended Medicines
 Display List Of Recommended Doctors

On Error:
 Show Alert With Error Message

5. FEATURES SECTION

Section: Why Choose Our Ayurvedic Predictor? Display 3
Features In Cards:
 - Ai-Powered Accuracy
 - Wide Range Of Remedies
 - Doctor Recommendations

6. FAQ SECTION

Section: Frequently Asked Questions For Each
Faq:
 Create Clickable Question
 Toggle Answer Visibility When Clicked

7. CONTACT SECTION

Section: Contact Us
Form With Fields:
 - Full Name
 - Email Address
 - Message
Submit Button: Send Message
(Form Action Is Placeholder - No Actual Backend Handling)

8. FOOTER

Display © 2025 Ayurvedic Assistant. All Rights Reserved.

9. JAVASCRIPT FUNCTIONS

Define Function To Handle "Predict" Button Click

 Validate Input

 Fetch Response From Backend Using Post Request If

Success:

 Update Result Section With Predictions Else:

 Show Error

Define Function Toggleanswer()

 Toggle Faq Answer Visibility When A Question Is Clicked

APP.PY

1. IMPORT NECESSARY LIBRARIES AND RELATED LIBRARIES

 Import Pandas For Data Handling

 Import Scikit-Learn For ML Model, Vectorization, And Training

 Import Cors For Cross-Origin Requests

 Import Random For Selecting Doctors

2. INITIALIZE FLASK APP AND ENABLE CORS

 Create Flask App Instance

 Enable Cors To Handle Cross-Origin Api Requests

3. LOAD AND PREPROCESS DATA

 Read 'Ayurvedic_Symptoms_Desc.Csv' Into Symptoms_Data

 Read 'Formulation-Indications.Csv' Into Indications_Data

 Convert Text Columns To Lowercase For Consistency:

 - Symptoms_Data['Description']

 - Indications_Data['Main Indications']

4. PREPARE DATA FOR SYMPTOM PREDICTION MODEL

Define Feature X As Descriptions
Define Target Y As Symptom Labels

Use Tf-Idf Vectorizer To Convert X Into Numerical Features
Split Data Into Training And Testing Sets (80/20)
Train Multinomial Naive Bayes Classifier On Training Set

5. DEFINE SAMPLE DOCTOR LIST

Create A List Of Doctor Dictionaries With Name And Phone Number

6. DEFINE FLASK ROUTES

6.1 HOME ROUTE (/)

Render Html Interface (Index.Html)

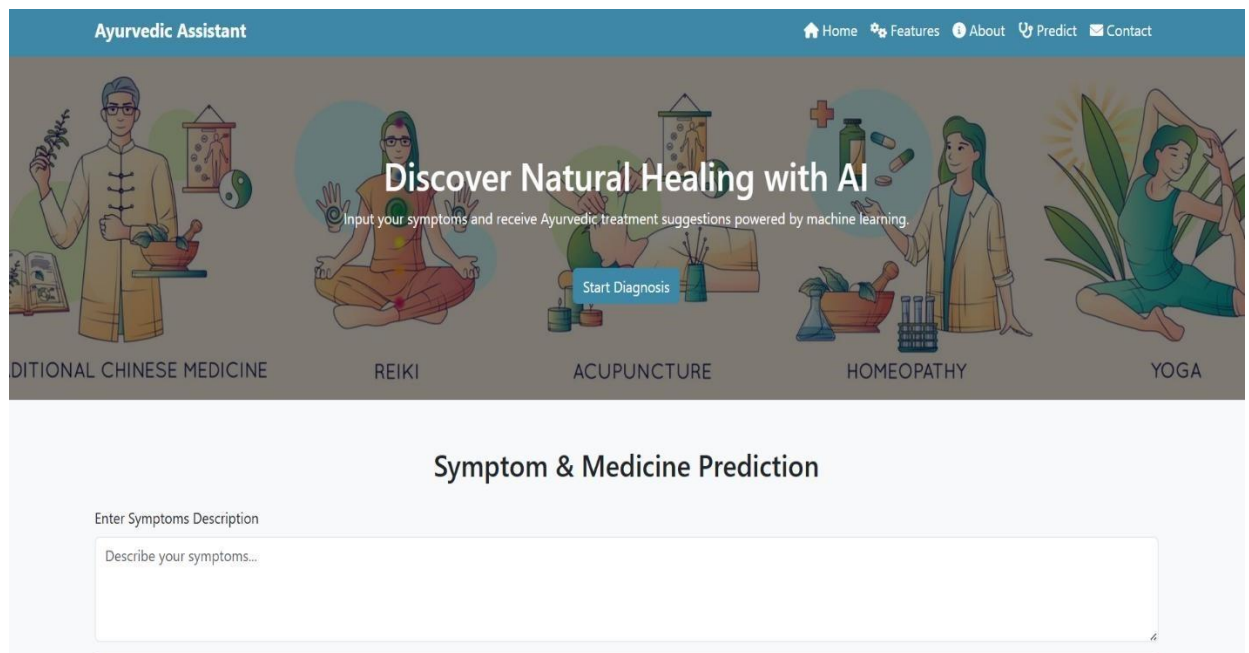
6.2 PREDICT ROUTE (/PREDICT, POST)

Receive Json Input From User (Contains 'Description' Of Symptoms)
Convert Input To Lowercase And Vectorize Using Tf-Idf
Predict The Symptom Using The Trained Model
Search Indications_Data For Formulations Whose 'Main Indications'
Contain The Predicted Symptom
Extract Top 5 Matching Medicines
Randomly Select 5 Doctors From Doctor List Return Json
Response Containing:
- Predicted_Symptom
- List Of Predicted_Medicines
- List Of Selected Doctors

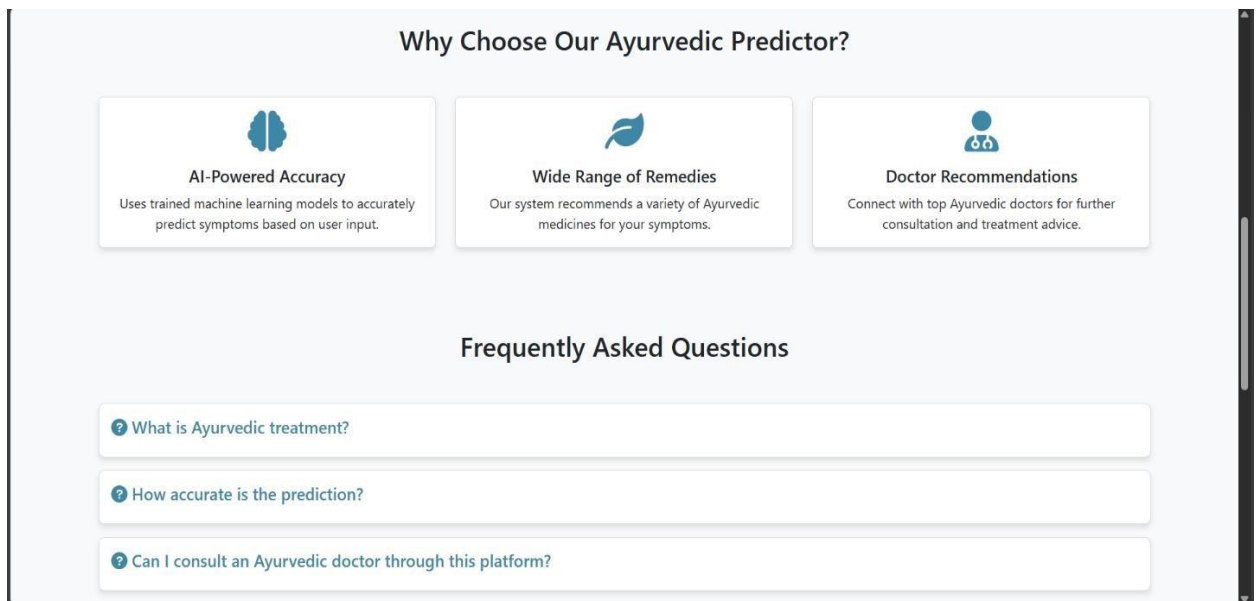
7. RUN FLASK APP

Start The Flask Server In Debug Mode

APPENDIX-B SCREENSHOTS



Home Page



Information Page

Contact Us

Full Name

Enter your name

Email Address

Enter your email

Message

Write your message here...

Send Message

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Contact Page

Enter Symptoms Description

head pain , fever

Predict

Predicted Symptom:

Shiiroroga

Recommended Medicines:

- Pathyadi Kvatha(Shadanga) Churna
- Pippali mula Churna
- Bhringaraja Tail
- Shadabindu Taila
- Akika Pishti

Recommended Doctors:

- Dr. Nitin Deshmukh - Phone: 7000000006
- Dr. Priya Kaur - Phone: 7000000007
- Dr. Manish Verma - Phone: 7000000010
- Dr. Anita Joshi - Phone: 7000000001
- Dr. Ravi Patel - Phone: 9123456780

Result Page

APPENDIX-C ENCLOSURES

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AYURSOFTAPP- Software That Suggests Drugs and Formulations for Disease/ Pharmacological Property Based on the Ayurvedic Classical Books/Repositories
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0% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

-  **0 AI-generated only 0%**
Likely AI-generated text from a large-language model.
-  **0 AI-generated text that was AI-paraphrased 0%**
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely AI generated as AI generated and AI paraphrased or likely AI generated and AI paraphrased writing as only AI generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.





3% Overall Similarity

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